

# Equation of State of the PDL Coherence Fluid: Resolution of OP-pressure

PDL Document D54

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## Abstract

The PDL coherence fluid, whose stress-energy tensor  $C_{\text{coh}}$  was fully derived in D48v3 [1], has a well-defined equation of state  $w = P/(\rho_{\text{coh}}c^2)$ . The question of whether this fluid behaves as dust, radiation, or dark energy has been open since D35 introduced the coherence source term [2]. This document resolves OP-pressure by assembling the four theorems established in D48v3 and D49 into a single coherent statement.

In the homogeneous ground state, the PDL coherence fluid satisfies  $w_{\text{mass}} = 0$ ,  $w_{\text{spin}} = 0$ , and  $w_{\text{orb}} = 0$ , each by a distinct unconditional theorem of axioms C1–C4. The leakage component has  $w_{\text{leak}} = -1$  intrinsically, carrying a weight  $\rho_{\Lambda}/\rho_{\text{coh}}(N) \sim 10^{-44}$  at coherence densities. The complete equation of state is

$$w(N) = -\frac{\rho_{\Lambda}}{\rho_{\text{coh}}(N)},$$

an unconditional theorem of C1–C4. At coherence densities the fluid is pressureless cold matter ( $|w| \sim 10^{-44}$ ); at cosmological scales the leakage density  $\rho_{\Lambda} = 5.835 \times 10^{-27} \text{ kg m}^{-3}$  corresponds to  $\Omega_{\Lambda}^{\text{PDL}} = \rho_{\Lambda}/\rho_{\text{crit}} = 0.6838$ , in agreement with the Planck 2020 value  $\Omega_{\Lambda}^{\text{obs}} = 0.685$  to 0.17%. Both regimes emerge from the same combinatorial structure, the proton quintuplet (24, 28, 930, 10087, 11017), without any parameter adjustment. All results are verified at 50-decimal precision (Python/mpmath).

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# 1 Introduction and context

## 1.1 A question open since D35

The PDL Einstein field equation, established in D35 [2], takes the form

$$\mathcal{G}[g_{\mu\nu}^{\text{eff}}] = \kappa_{\text{PDL}}(N) C_{\text{coh}}, \quad (1)$$

where  $\kappa_{\text{PDL}}(N) = 8\pi\sigma(N)G_{\text{PDL}}/c^4$  (with  $\sigma(N) = 1 - (1 - \kappa)^N$  and  $\kappa = 310\varphi/11017$ , an unconditional theorem of C1–C4 since D42 [3]) and  $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$  is the coherence stress-energy tensor. For any source in the Einstein equation, the equation of state  $w = P/(\rho c^2)$  is a fundamental characterisation: it determines whether the fluid decelerates or accelerates cosmic expansion, whether it clusters gravitationally, and how its energy density evolves with the scale factor.

When D35 was written, the components of  $C_{\text{coh}}$  were only schematically identified. Documents D48v3 [1] and D49 [4] completed the derivation of all four components from C1–C4. The present document assembles those results to close OP-pressure.

## 1.2 What makes this resolution possible now

Three developments converged to make D54 possible. First, D48v3 [1] corrected four dimensional errors present in earlier versions and derived  $C_{\mu\nu}^{(\text{spin})} = 0$  in the homogeneous ground state as an unconditional theorem (OP-spin resolved), placing all four components of  $C_{\text{coh}}$  in consistent units of  $\text{J m}^{-3}$ . Second, D49 [4] proved that axiom C4 forces the London gauge  $\nabla\phi = 0$  over  $V_C$ , giving  $j^i = 0$  in the ground state and therefore  $C_{\mu\nu}^{(\text{orb})} = 0$ . Third, D51–D53 [5, 6, 7] completed the derivation of  $\Lambda_{\text{PDL}}$ , giving  $C_{\mu\nu}^{(\text{leak})}$  a fully determined numerical value verified to 0.41 ppm.

## 1.3 The open problem resolved

**Resolved Problem 1** (OP-pressure — equation of state of the PDL coherence fluid). *Derive  $w = P/(\rho_{\text{coh}}c^2)$  of the PDL coherence fluid from C1–C4.*

*Resolution (this document). In the homogeneous ground state:  $w_{\text{mass}} = w_{\text{spin}} = w_{\text{orb}} = 0$  (three unconditional theorems);  $w_{\text{leak}} = -1$  intrinsically with weight  $\rho_{\Lambda}/\rho_{\text{coh}}(N) \sim 10^{-44}$  at coherence densities. The complete equation of state  $w(N) = -\rho_{\Lambda}/\rho_{\text{coh}}(N)$  is an unconditional theorem of C1–C4 (Theorem 4.1).*

## 2 Prerequisites

**Theorem 2.1** (Coherence density [1]).  $\rho_{\text{coh}}(N) = \sigma(N) R_{\text{surf}} m_p / (V_C R_{\text{tot}})$ , where  $V_C = (4\pi/3)\lambda_C^3$ ,  $\lambda_C = \hbar/(m_p c)$ . Unconditional theorem of C1–C4, verified to  $< 0.013\%$  against D48v3.

**Theorem 2.2** (Mass component [1]).  $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}(N) c^2 u_{\mu} u_{\nu}$ . Units:  $\text{J m}^{-3}$ . Unconditional theorem.

**Theorem 2.3** (Spin component and OP-spin [1]).  $C_{\mu\nu}^{(\text{spin})} = 0$  in the homogeneous ground state. From  $\langle S^i \rangle = 0$  (Lemma 5.1 of D48v3, 8 coherent configurations of  $K_4$ ) and  $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$  (Lemma 5.2), the Weyssenhoff–Raabe traceless part vanishes exactly. Unconditional theorem (OP-spin resolved).

**Theorem 2.4** (London gauge and orbital component [4, 1]). *C4 forces  $\nabla\phi = 0$  over  $V_C$  (D49, Theorem 1, 768 configurations). Hence  $j^i = 0$  and  $C_{\mu\nu}^{(\text{orb})} = m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}} = 0$ . Unconditional theorem.*

**Theorem 2.5** (Leakage component [1, 5, 6]).  *$C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}} c^4 / (8\pi G_{\text{PDL}} \sigma(N)) g_{\mu\nu}^{\text{eff}}$ , with  $\Lambda_{\text{PDL}} = \mathcal{C} \eta_L^{18} / \lambda_C^2$  (D51+D52, D53 [7]). Units:  $\text{J m}^{-3}$ . Verified:  $\kappa_{\text{PDL}}(N) \cdot C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}}$  to  $< 10^{-48}\%$  for all  $N$ , independent of  $N$ . Unconditional theorem.*

### 3 The equation of state: component by component

#### 3.1 Mass component: $w_{\text{mass}} = 0$

**Theorem 3.1.**  $w_{\text{mass}} = P_{\text{mass}} / (\rho_{\text{coh}} c^2) = 0$  exactly.

*Proof.* By Theorem 2.2,  $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}} c^2 u_\mu u_\nu$ . For pressureless dust [8],  $T_{ij} = 0$  in the comoving frame and  $P_{\text{mass}} = 0$ . The kinetic stress appears entirely in  $C_{\mu\nu}^{(\text{orb})}$  via the Madelung decomposition (D48v3, proof of Theorem 7.1).  $\square$

#### 3.2 Spin component: $w_{\text{spin}} = 0$

**Theorem 3.2.**  $w_{\text{spin}} = 0$  exactly in the homogeneous ground state.

*Proof.* By Theorem 2.3,  $C_{\mu\nu}^{(\text{spin})} = 0$ , so  $P_{\text{spin}} = \frac{1}{3} \text{tr}(C_{ij}^{(\text{spin})}) = 0$ .  $\square$

**Remark 3.3.** *The mechanism is the perfect isotropy of  $\langle S^i S^j \rangle_8 = \frac{1}{4} \delta^{ij}$  over the  $K_4$  configurations, from which the Weyssenhoff–Raabe [9] shear stress vanishes identically. In the inhomogeneous regime with inter-cell scale  $L \gg \lambda_C$ , the spin pressure is suppressed by  $(\lambda_C/L)^2 \rightarrow 0$ .*

#### 3.3 Orbital component: $w_{\text{orb}} = 0$

**Theorem 3.4.**  $w_{\text{orb}} = 0$  exactly in the homogeneous ground state without external electromagnetic field.

*Proof.* By Theorem 2.4, the London gauge  $\nabla\phi = 0$  holds over  $V_C$ . The PDL probability current  $j^i = (\hbar/m_p) \text{Im}(\psi^* \partial^i \psi) = |\psi|^2 \partial^i \phi / m_p = 0$ . Therefore  $C_{ij}^{(\text{orb})} = 0$  and  $P_{\text{orb}} = 0$ .  $\square$

#### 3.4 Leakage component: $w_{\text{leak}} = -1$

**Theorem 3.5.** *The leakage component has intrinsic equation of state  $w_{\text{leak}} = -1$ .*

*Proof.* By Theorem 2.5,  $C_{\mu\nu}^{(\text{leak})} \propto g_{\mu\nu}^{\text{eff}}$ . A term proportional to  $g_{\mu\nu}^{\text{eff}}$  in the stress-energy tensor corresponds to  $P_\Lambda = -\rho_\Lambda c^2$  with  $\rho_\Lambda = \Lambda_{\text{PDL}} c^2 / (8\pi G_{\text{PDL}})$ , hence  $w_{\text{leak}} = -1$ .  $\square$

**Remark 3.6** (Numerical value of  $\rho_\Lambda$ ). From  $\Lambda_{\text{PDL}} = 1.089 \times 10^{-52} \text{ m}^{-2}$  (D51–D53, verified to 0.41 ppm):

$$\rho_\Lambda = \frac{\Lambda_{\text{PDL}} c^2}{8\pi G_{\text{PDL}}} = 5.835 \times 10^{-27} \text{ kg m}^{-3}. \quad (2)$$

*The corresponding PDL dark energy fraction is*

$$\Omega_\Lambda^{\text{PDL}} = \frac{\rho_\Lambda}{\rho_{\text{crit}}} = \frac{5.835 \times 10^{-27}}{8.533 \times 10^{-27}} = 0.6838, \quad (3)$$

where  $\rho_{\text{crit}} = 3H_0^2/(8\pi G) = 8.533 \times 10^{-27} \text{ kg m}^{-3}$  for  $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (Planck 2020 [10]). The observed value is  $\Omega_{\Lambda}^{\text{obs}} = 0.685$ , giving a discrepancy of 0.17% — within the current uncertainty on  $H_0$  and  $\Lambda_{\text{PDL}}$ .

## 4 The complete equation of state

**Theorem 4.1** (Complete equation of state of the PDL coherence fluid). *In the homogeneous ground state, the equation of state of the PDL coherence fluid is*

$$\boxed{w(N) = -\frac{\rho_{\Lambda}}{\rho_{\text{coh}}(N)}}, \quad (4)$$

where  $\rho_{\Lambda} = \Lambda_{\text{PDL}} c^2/(8\pi G_{\text{PDL}})$  and  $\rho_{\text{coh}}(N) = \sigma(N) R_{\text{surf}} m_p/(V_C R_{\text{tot}})$ . Every quantity in (4) is an unconditional theorem of C1–C4. There is no free parameter.

*Proof.* Theorems 3.1, 3.2, 3.4 give  $w_{\text{mass}} = w_{\text{spin}} = w_{\text{orb}} = 0$ . Theorem 3.5 gives  $w_{\text{leak}} = -\rho_{\Lambda}/\rho_{\text{coh}}(N)$ . Summing:  $w(N) = -\rho_{\Lambda}/\rho_{\text{coh}}(N)$ .  $\square$

**Corollary 4.2** (Two physical regimes). *Equation (4) exhibits two physically distinct regimes.*

**(a) Coherence and nuclear densities** ( $N = 1\text{--}200$ ,  $\rho_{\text{coh}} \sim 10^{17}\text{--}10^{18} \text{ kg m}^{-3}$ ):  $|w(N)| \sim 10^{-44}$ . The fluid is pressureless cold matter, indistinguishable in equation of state from cold dark matter.

**(b) Cosmological scales:** the leakage density  $\rho_{\Lambda} = 5.835 \times 10^{-27} \text{ kg m}^{-3}$  is comparable to the critical density  $\rho_{\text{crit}} = 8.533 \times 10^{-27} \text{ kg m}^{-3}$ , giving  $\Omega_{\Lambda}^{\text{PDL}} = 0.6838$ , in agreement with  $\Omega_{\Lambda}^{\text{obs}} = 0.685$  (Planck 2020) to 0.17%.

**Remark 4.3** (A single fluid, two regimes). *Corollary 4.2 shows that the PDL framework does not postulate two distinct dark components. A single coherence fluid, characterised entirely by C1–C4 and the quintuplet (24, 28, 930, 10087, 11017), interpolates between pressureless cold matter at sub-nuclear scales and a cosmological-constant contribution at cosmic scales. The transition is governed by  $\rho_{\Lambda}/\rho_{\text{coh}}(N)$ , itself a theorem of C1–C4. Whether the coherence fluid is to be identified with the observed dark matter component of the universe is a cosmological question that lies beyond the scope of the present document (OP-DM, Section 8).*

## 5 Numerical verification

All computations use Python/mpmath at 50-decimal precision. The complete script (file D54\_OP\_pressure\_final.py) is provided as supplementary material.

### 5.1 Spin averages (Lemmas 5.1–5.2 of D48v3)

$$\langle S^x \rangle = \langle S^y \rangle = \langle S^z \rangle = 0.00 \times 10^0, \quad (5)$$

$$\max_{i,j} \left| \langle S^i S^j \rangle - \frac{1}{4} \delta^{ij} \right| = 0.00 \times 10^0, \quad \max_{i,j} \left| \langle S^i S^j \rangle_{\text{tl}} \right| = 0.00 \times 10^0. \quad (6)$$

All exact at machine precision, confirming Theorems 2.3 and 3.2.

## 5.2 London gauge (D49 Lemma 1)

Violation fractions  $\nu_k = \frac{1}{4} \|\psi_1 - T^k \psi_1\|^2$ :

| $k$     | 0      | 1      | 2      | 3      |
|---------|--------|--------|--------|--------|
| $\nu_k$ | 0.0000 | 0.5000 | 1.0000 | 0.5000 |

Unique minimum at  $k = 0$ ; 32/32 consistency checks passed. Confirms Theorems 2.4 and 3.4.

## 5.3 Leakage term: $\kappa_{\text{PDL}}(N) \cdot C^{(\text{leak})} = -\Lambda_{\text{PDL}}$

Table 1: Verification of  $\kappa_{\text{PDL}}(N) \cdot C^{(\text{leak})} = -\Lambda_{\text{PDL}}$  and  $\log_{10}(\rho_{\Lambda}/\rho_{\text{coh}}(N))$ .

| $N$ | $\rho_{\text{coh}}(N) [\text{kg m}^{-3}]$ | $\kappa_{\text{PDL}} \cdot C^{(\text{leak})} [\text{m}^{-2}]$ | $\Delta/\Lambda_{\text{PDL}}$ | $\log_{10}(\rho_{\Lambda}/\rho_{\text{coh}})$ |
|-----|-------------------------------------------|---------------------------------------------------------------|-------------------------------|-----------------------------------------------|
| 1   | $8.898 \times 10^{16}$                    | $-1.089 \times 10^{-52}$                                      | 0.0%                          | -43.18                                        |
| 5   | $4.062 \times 10^{17}$                    | $-1.089 \times 10^{-52}$                                      | 0.0%                          | -43.84                                        |
| 10  | $7.280 \times 10^{17}$                    | $-1.089 \times 10^{-52}$                                      | 0.0%                          | -44.10                                        |
| 40  | $1.651 \times 10^{18}$                    | $-1.089 \times 10^{-52}$                                      | 0.0%                          | -44.45                                        |
| 120 | $1.947 \times 10^{18}$                    | $-1.089 \times 10^{-52}$                                      | $2.7 \times 10^{-49}\%$       | -44.52                                        |
| 200 | $1.954 \times 10^{18}$                    | $-1.089 \times 10^{-52}$                                      | 0.0%                          | -44.53                                        |

$\kappa_{\text{PDL}} \cdot C^{(\text{leak})}$  is constant across all  $N$  to  $< 10^{-48}\%$ , confirming the independence of  $\Lambda_{\text{PDL}}$  on  $N$ .

## 5.4 Cosmological comparison

The PDL dark energy density  $\rho_{\Lambda} = 5.835 \times 10^{-27} \text{ kg m}^{-3}$  compared to the cosmological critical density  $\rho_{\text{crit}} = 3H_0^2/(8\pi G) = 8.533 \times 10^{-27} \text{ kg m}^{-3}$  (Planck 2020,  $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  [10]):

$$\Omega_{\Lambda}^{\text{PDL}} = \frac{\rho_{\Lambda}}{\rho_{\text{crit}}} = \frac{5.835 \times 10^{-27}}{8.533 \times 10^{-27}} = 0.6838, \quad (7)$$

to be compared with  $\Omega_{\Lambda}^{\text{obs}} = 0.685$  (Planck 2020). The discrepancy is 0.17%, well within the uncertainty on  $H_0$  and the 0.41 ppm precision of  $\Lambda_{\text{PDL}}$  (D53 [7]).

# 6 Physical interpretation

## 6.1 Cold dark matter at coherence scales

At the scale of individual nucleons and nuclei ( $N = 1\text{--}200$ ,  $\rho_{\text{coh}} \sim 10^{17}\text{--}10^{18} \text{ kg m}^{-3}$ ), equation (4) gives  $|w(N)| \sim 10^{-44}$ , indistinguishable from zero. The PDL coherence fluid is pressureless cold matter. Three independent mechanisms enforce this: the dust structure of  $C^{(\text{mass})}$ ; the isotropic spin distribution over  $K_4$ ; and the London gauge forced by C4. There is no velocity dispersion and no free-streaming length.

## 6.2 Dark energy at cosmological scales

At cosmological scales, the leakage density  $\rho_\Lambda = 5.835 \times 10^{-27} \text{ kg m}^{-3}$  is comparable to the critical density. The PDL dark energy fraction  $\Omega_\Lambda^{\text{PDL}} = 0.6838$  agrees with the Planck 2020 measurement  $\Omega_\Lambda^{\text{obs}} = 0.685$  to 0.17%. This level of agreement is remarkable:  $\Lambda_{\text{PDL}}$  was derived from the combinatorial structure of C1–C4 and the proton quintuplet alone, with no cosmological input.

## 6.3 The unifying perspective

The PDL coherence fluid provides a unified description of two phenomena that, in the  $\Lambda$ CDM concordance model, are attributed to two distinct and unrelated postulates: cold dark matter and the cosmological constant. In the PDL framework, both emerge from the same object  $C_{\text{coh}}$  under the same four axioms. The transition between the two regimes is mediated entirely by the ratio  $\rho_\Lambda/\rho_{\text{coh}}(N)$ , itself an unconditional theorem. Whether this unification is to be identified with the observed dark sector is a cosmological question that requires a dedicated analysis (OP-DM, Section 8); D54 establishes only the equation-of-state theorem and records the numerical coincidence at 0.17%.

## 7 Epistemic status

Table 2: Epistemic status of the results of D54. All numerical verifications at 50-decimal precision (Python/mpmath).

| Result                                                                                  | Status          | Source              |
|-----------------------------------------------------------------------------------------|-----------------|---------------------|
| $w_{\text{mass}} = 0$                                                                   | <b>Theorem</b>  | D48v3 Thm 4.1       |
| $w_{\text{spin}} = 0$ (ground state)                                                    | <b>Theorem</b>  | D48v3 Thm 5.3       |
| $\langle S^i \rangle = 0$ (exact, 8 configs)                                            | <b>Verified</b> | D48v3 Lem. 5.1      |
| $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ (exact)                             | <b>Verified</b> | D48v3 Lem. 5.2      |
| $w_{\text{orb}} = 0$ (ground state)                                                     | <b>Theorem</b>  | D49 Thm 1, D48v3    |
| $\nu_k = (0, \frac{1}{2}, 1, \frac{1}{2})$ (32/32)                                      | <b>Verified</b> | D49 Lem. 1          |
| $w_{\text{leak}} = -1$ (intrinsic)                                                      | <b>Theorem</b>  | D48v3 Prop. 8.1     |
| $\kappa_{\text{PDL}}(N) \cdot C^{(\text{leak})} = -\Lambda_{\text{PDL}} (< 10^{-48}\%)$ | <b>Verified</b> | Python/mpmath       |
| $\rho_\Lambda = 5.835 \times 10^{-27} \text{ kg m}^{-3}$                                | <b>Verified</b> | D51–D53             |
| $\rho_\Lambda/\rho_{\text{coh}}(N) \sim 10^{-44}$ (all $N$ )                            | <b>Verified</b> | Python/mpmath       |
| $\Omega_\Lambda^{\text{PDL}} = 0.6838; \Omega_\Lambda^{\text{obs}} = 0.685$             | <b>Verified</b> | Planck 2020         |
| $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$                                             | <b>Theorem</b>  | Thm 4.1             |
| $ w  \sim 10^{-44}$ at coherence densities                                              | <b>Theorem</b>  | Cor. 4.2(a)         |
| $\Omega_\Lambda^{\text{PDL}}$ agrees with obs. to 0.17%                                 | <b>Verified</b> | Cor. 4.2(b)         |
| OP-pressure resolved                                                                    | <b>Theorem</b>  | D54 (this document) |
| Dark matter identification (OP-DM)                                                      | <b>Open</b>     | D24, D27, D35, D54  |
| Inhomogeneous regime ( $L \sim \lambda_C$ )                                             | <b>Open</b>     | D48v3, D49, D54     |

## 8 Open problems

**Open Problem 1** (OP-DM — Identification of the coherence fluid with observed dark matter). *Theorem 4.1 proves that the PDL coherence fluid has  $w \approx 0$  at nuclear and sub-nuclear*



densities, identical in equation of state to cold dark matter. Whether the fluid can be identified with the observed dark matter component of the universe — i.e. whether  $\rho_{\text{coh}}(N)$  integrated over cosmological scales accounts for the observed matter density  $\Omega_m \rho_{\text{crit}}$  — requires a dedicated cosmological analysis involving the density profile of coherence cells, their clustering properties, and their gravitational lensing signature. **Priority:** medium. **Entry point:** D24, D27, D35, D54.

**Open Problem 2** (OP-inhomogeneous — Equation of state beyond the ground state). Theorem 4.1 applies to the homogeneous ground state. In the inhomogeneous regime (spin orientations varying on scale  $L \gtrsim \lambda_C$ , or in the presence of an external vector potential),  $P_{\text{orb}}$  and  $P_{\text{spin}}$  are non-zero. Deriving the full equation of state in that regime from C1–C4 requires specifying the inter-cell quantum state distribution. **Priority:** medium. **Entry point:** D48v3, D49, D54.

## 9 Summary

The PDL coherence fluid has a well-defined equation of state derivable from axioms C1–C4 alone. Four components, four theorems. The mass component is pressureless dust:  $w_{\text{mass}} = 0$ . The spin component vanishes by the perfect isotropy of the  $K_4$  spin distribution:  $w_{\text{spin}} = 0$ . The orbital component vanishes because C4 forces the London gauge over  $V_C$ :  $w_{\text{orb}} = 0$ . The leakage component is a cosmological constant:  $w_{\text{leak}} = -1$ .

The complete equation of state  $w(N) = -\rho_{\Lambda}/\rho_{\text{coh}}(N)$  is an unconditional theorem of C1–C4 with no free parameter. At coherence densities,  $|w| \sim 10^{-44}$  (pressureless cold matter); at cosmological scales, the PDL dark energy fraction  $\Omega_{\Lambda}^{\text{PDL}} = 0.6838$  agrees with the Planck 2020 measurement  $\Omega_{\Lambda}^{\text{obs}} = 0.685$  to 0.17%.

The two dark components of the concordance cosmology arise from a single combinatorial fluid governed by the proton quintuplet (24, 28, 930, 10087, 11017) and axioms C1–C4, distinguished only by the density regime in which they are observed.



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